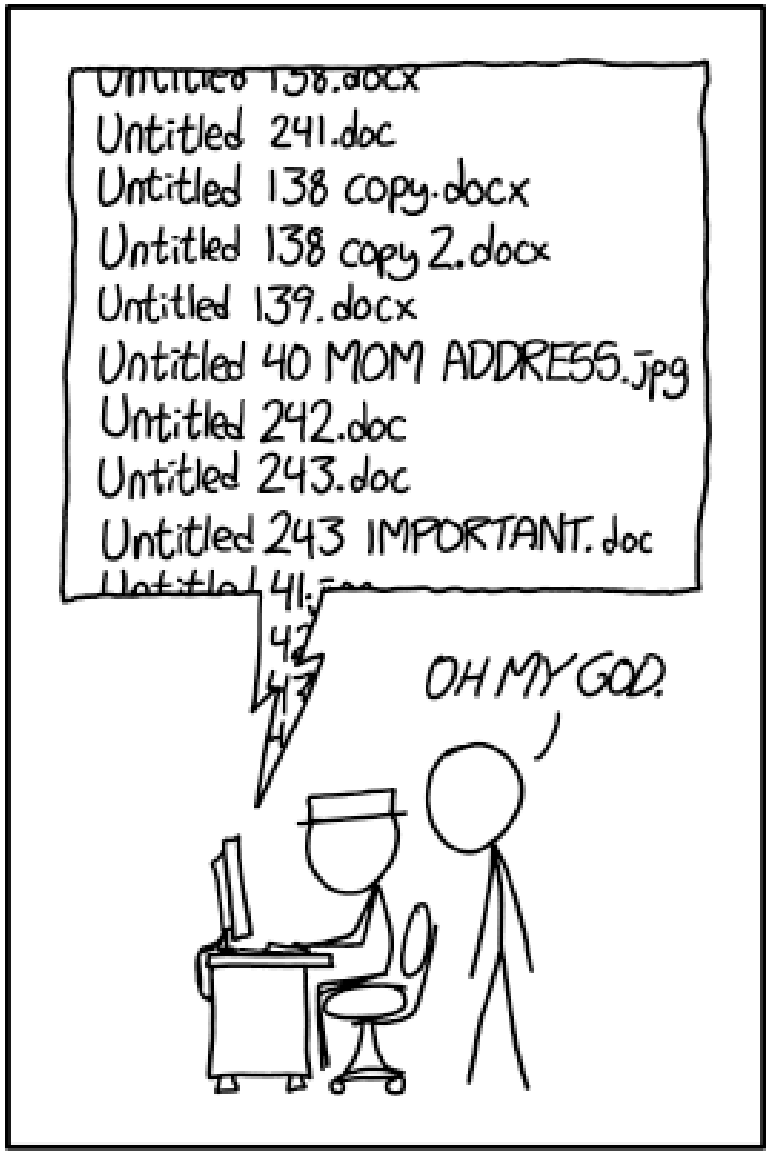
Using Git and GitHub

Tecnología de Videojuegos

University of Alcala

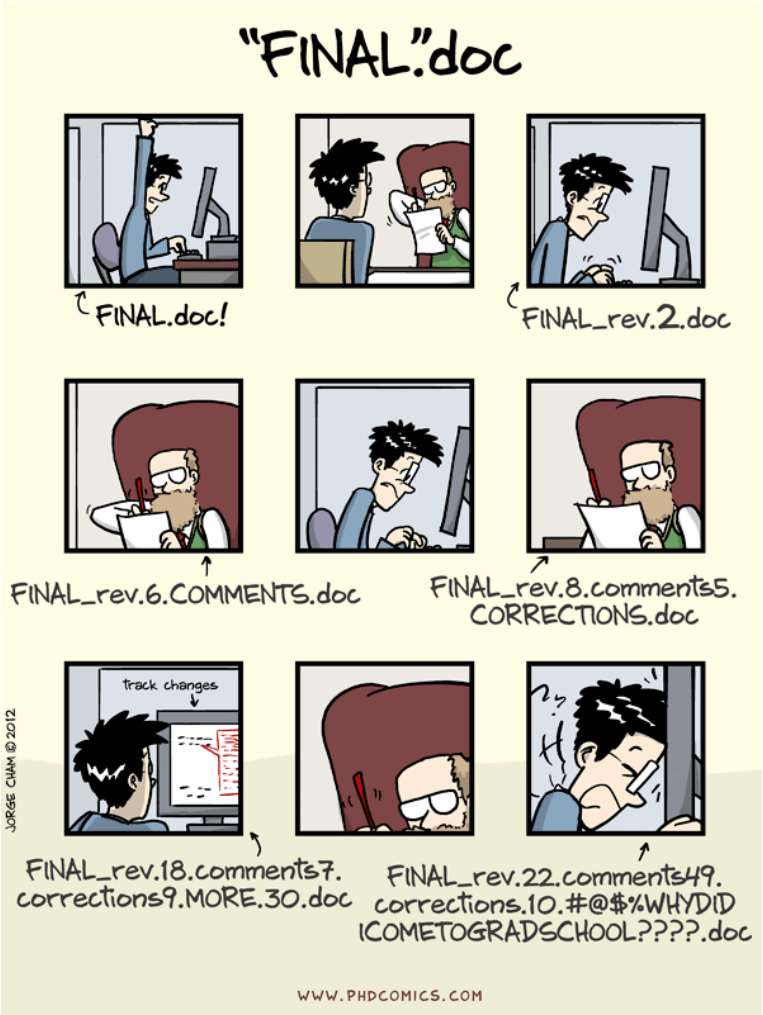
Table of Contents

1. Version control
   - Motivation
2. Git
   - What is Git?
   - Git sites
   - Motivation
   - Git vs. SVN
3. Using Git
   - Conflicts
   - Commits
   - Branches
   - Tags
   - Good practices
4. GitHub
   - Features
   - README
   - Markdown



PROTIP: NEVER LOOK IN SOMEONE ELSE'S DOCUMENTS FOLDER.

(Source)



(Source)

Version control

Introduction

Version control systems

Version control systems (VCS) keep track of changes to source code. Allows multiple people to edit a project in a predictable manner.

Main open source VCS

- 1982 RCS
- 1990 CVS
- 2000 Subversion
- 2005 Git/Mercurial

There are many proprietary ones but `Git` is now the most popular one by far.
All software should be under a version control system, if not, it ain't software!

Git

What is Git?

Git is an open source distributed version control system, created by Linus Torvald.
`https://git-scm.com/`
(Interactive tutorial)

Git

Git sites

It is easier to start with free hosting sites instead of maintaining your own server.

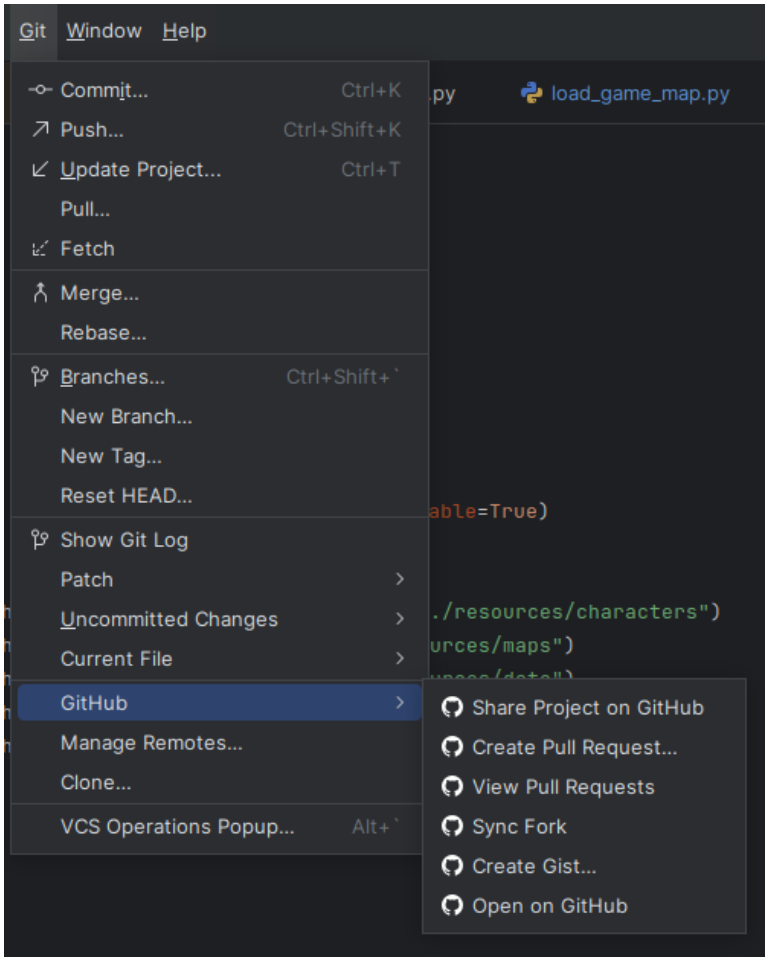
- GitHub: public repositories (as many as you want), but private ones are not free (except for academia). It is now part of Microsoft
- Bitbucket: allow us to keep private repositories limiting the number of collaborators.
- GitLab: both public and private without limitations. It is becoming more popular.
- Others ...

It is typically used as central repository:

- from which everyone pulls other people's changes
- to which everyone pushes changes they have made

Git vs. SVN

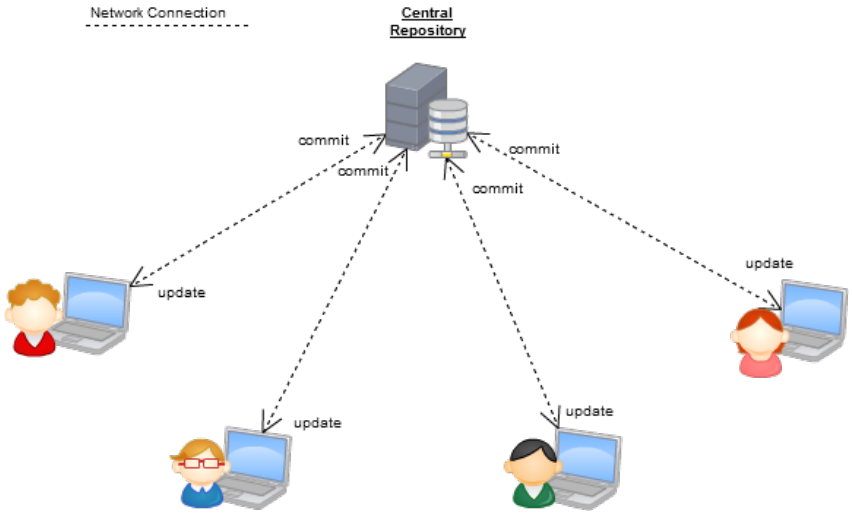
Motivation



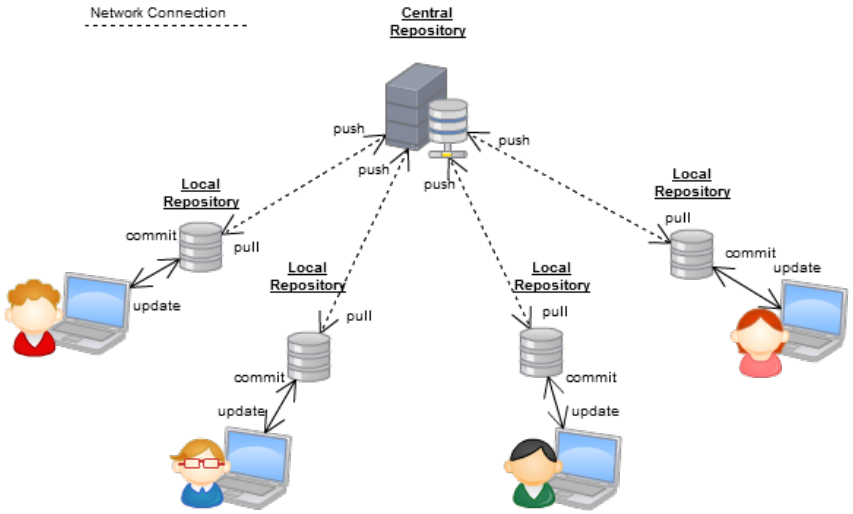
Git

Git vs. SVN (I)

Centralized (SVN)



Distributed (Git)



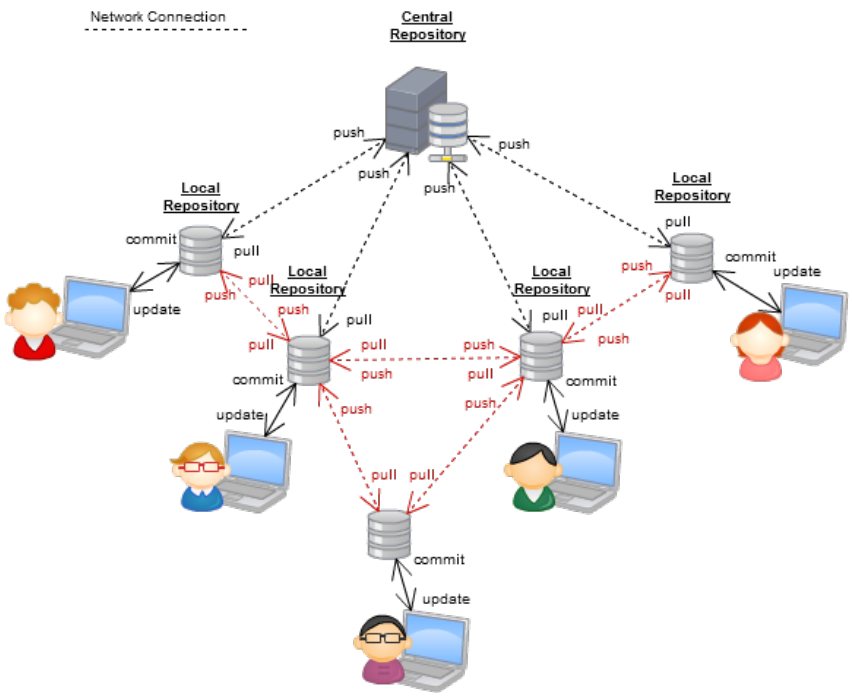
(Source)

Disclaimer: Do not pay attention to the labels of these diagrams

Git

Git vs. SVN (II)

Fully distributed (Git)



(Source)

Key Git concepts to know

- `repository` (local or not)
- `clone`
- `commit`, `push`
- `pull`, `fetch`
- `remote`, `origin`
- `merge`

Using Git

Conflicts

If two people both modify the same file, the first to push *wins*. The second person will have to pull and merge before pushing.

- Changes in different parts of a file are automatically merged
- Changes in the same part of a file cause conflicts
  - Select either your changes or remote, or a mix of the two

Using Git

Commits

Each commit has ...

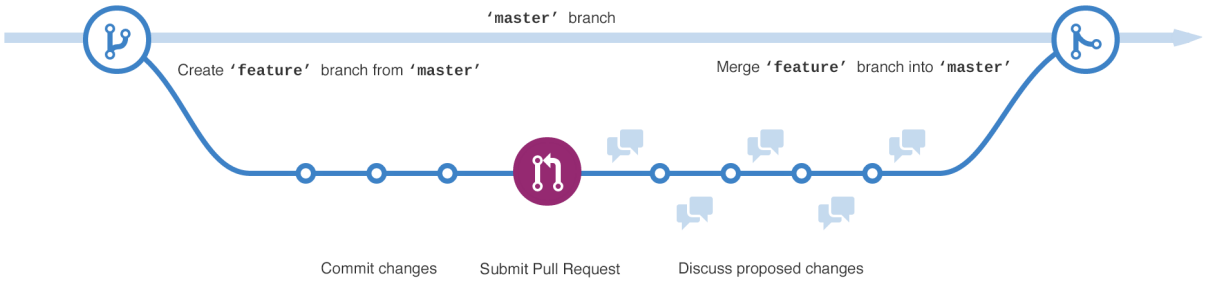
- ... an author
- ... a comment: "Fix deprecated py36 black option"
- ... a date
- ... an ID (or *hash*)

```
commit 44161dde6ea234f8cb997644f8e187123c3cc4af
Author: David <foo@foo.com>
Date:   Fri Mar 9 14:57:32 2018 +0100

    Issue with syntax highlighting solved
```

Using Git

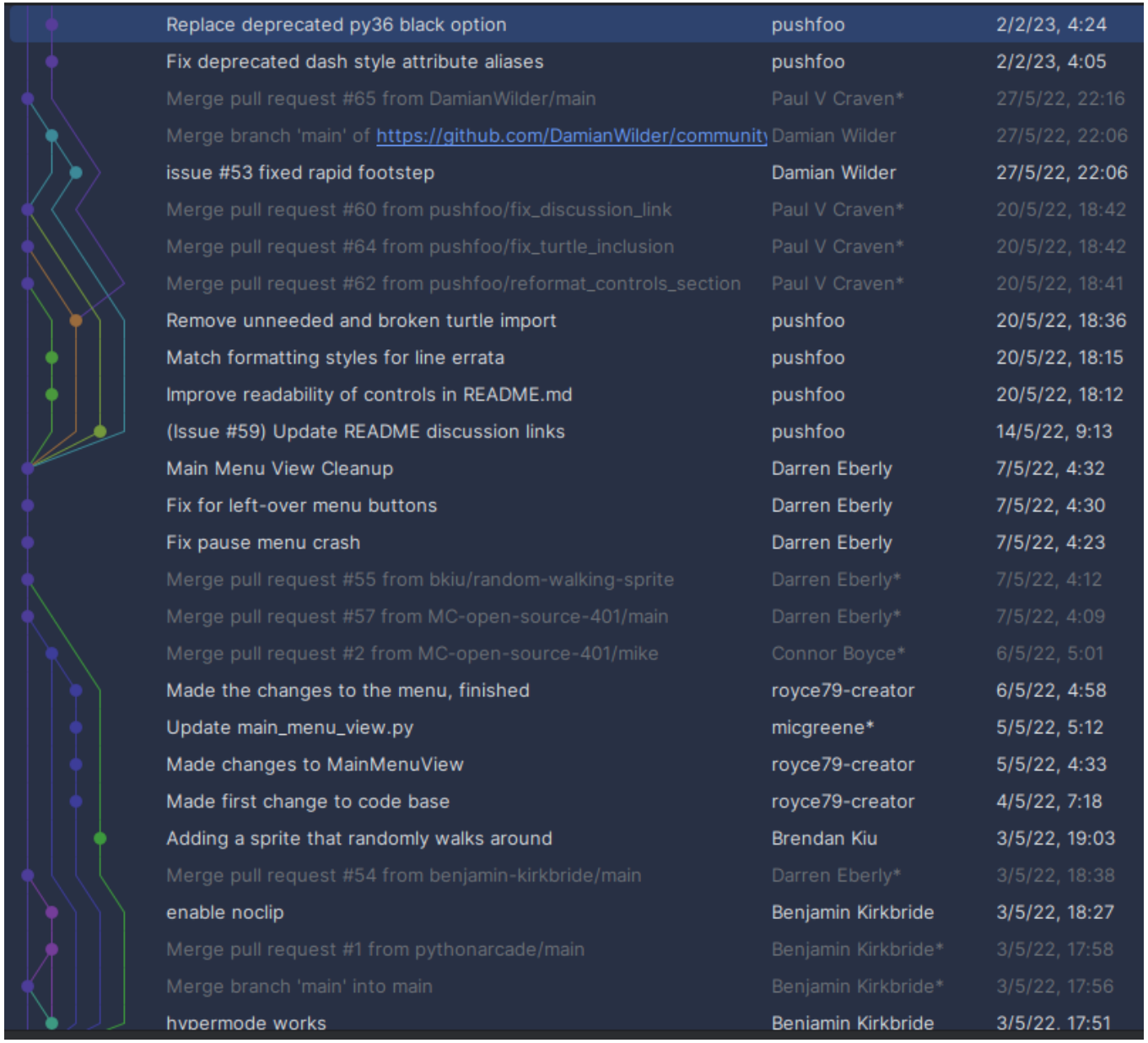
Branches (I)



Branches are used extensively (e.g. some like feature branches).

- A repository (local and remote) can have explicit branches
- The default branch is called master
- A merge is a fusion between two branches

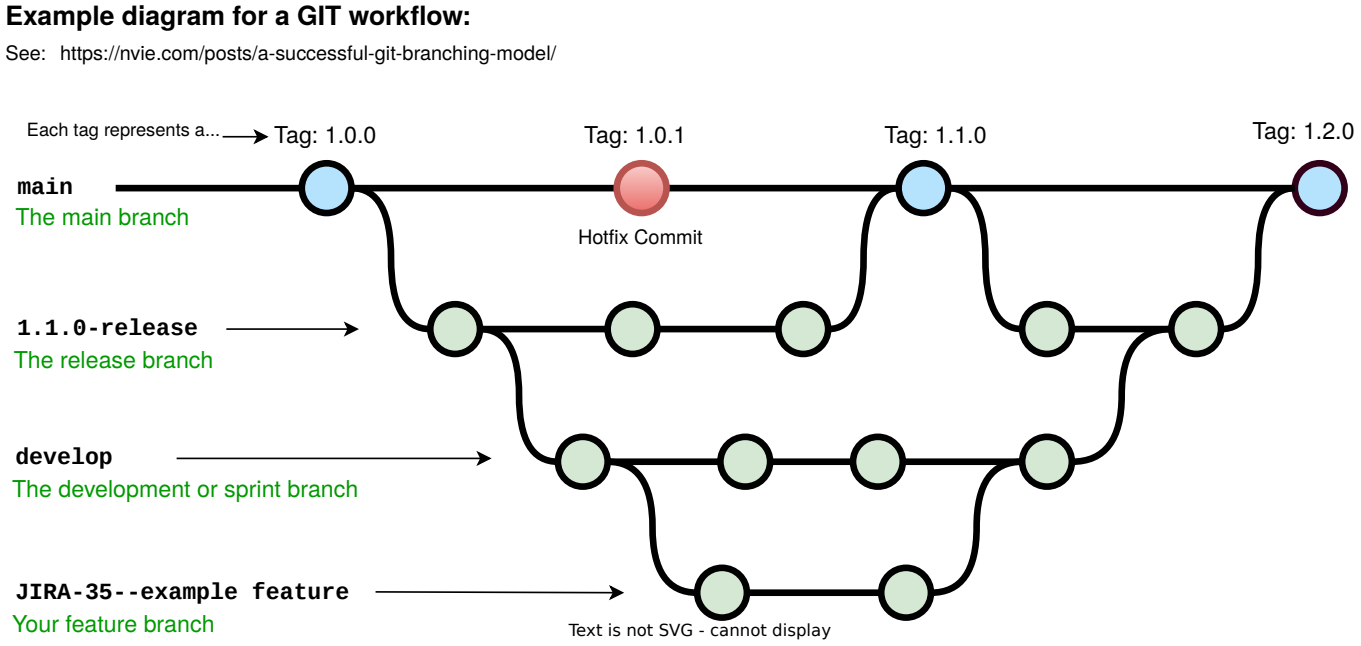
Do no use branches in the project!



Replace deprecated py36 black option	pushfoo	2/2/23, 4:24
Fix deprecated dash style attribute aliases	pushfoo	2/2/23, 4:05
Merge pull request #65 from DamianWilder/main	Paul V Craven*	27/5/22, 22:16
Merge branch 'main' of https://github.com/DamianWilder/community	Damian Wilder	27/5/22, 22:06
issue #53 fixed rapid footstep	Damian Wilder	27/5/22, 22:06
Merge pull request #60 from pushfoo/fix_discussion_link	Paul V Craven*	20/5/22, 18:42
Merge pull request #64 from pushfoo/fix_turtle_inclusion	Paul V Craven*	20/5/22, 18:42
Merge pull request #62 from pushfoo/reformat_controls_section	Paul V Craven*	20/5/22, 18:41
Remove unneeded and broken turtle import	pushfoo	20/5/22, 18:36
Match formatting styles for line errata	pushfoo	20/5/22, 18:15
Improve readability of controls in README.md	pushfoo	20/5/22, 18:12
(Issue #59) Update README discussion links	pushfoo	14/5/22, 9:13
Main Menu View Cleanup	Darren Eberly	7/5/22, 4:32
Fix for left-over menu buttons	Darren Eberly	7/5/22, 4:30
Fix pause menu crash	Darren Eberly	7/5/22, 4:23
Merge pull request #55 from bkiu/random-walking-sprite	Darren Eberly*	7/5/22, 4:12
Merge pull request #57 from MC-open-source-401/main	Darren Eberly*	7/5/22, 4:09
Merge pull request #2 from MC-open-source-401/mike	Connor Boyce*	6/5/22, 5:01
Made the changes to the menu, finished	royce79-creator	6/5/22, 4:58
Update main_menu_view.py	micgreene*	5/5/22, 5:12
Made changes to MainMenuView	royce79-creator	5/5/22, 4:33
Made first change to code base	royce79-creator	4/5/22, 7:18
Adding a sprite that randomly walks around	Brendan Kiu	3/5/22, 19:03
Merge pull request #54 from benjamin-kirkbride/main	Darren Eberly*	3/5/22, 18:38
enable noclip	Benjamin Kirkbride	3/5/22, 18:27
Merge pull request #1 from pythonarcade/main	Benjamin Kirkbride*	3/5/22, 17:58
Merge branch 'main' into main	Benjamin Kirkbride*	3/5/22, 17:56
hypermode works	Benjamin Kirkbride	3/5/22, 17:51

Using Git

Tags



A tag is a pointer to a specific point in the repository history

- Tags usually have names (e.g. "v1.1")
- Widely used to keep and publish software releases

Using Git

Good practices

Learn on the job: the best way to learn it is by using it.

Best practices

- Regularly push and pull (at least daily, in general)
- ⇒ Test before pushing! ⇐
- Don't push half-baked changes
- Don't pull if you're in the middle of a task
- Never commit temporal/intermediate files
- Keep commit descriptions short and informative
- *The master must be a clean and functional version of the project*

Remember: Git never overwrites local changes without an explicit order

- .. even with a `git pull`

GitHub

Features

Free Git hosting provider

- Free public repositories

Added value features

- Social network
- Collaborative tools
- Repository browser
- Pull requests
- Issue tracking
- Web hosting
- Markdown integration
- Organizations

GitHub

Key concepts

Key GitHub concepts to know

- `Pull request`
- `Fork`

GitHub

README

Special file: README.md

- Contains information about the project
- Automatically visualized
- md means Markdown

Markdown (I)

Markdown: Trivial markup

- Simple
- Very simple
- Extremely simple
- Did I say it's simple?

VERY powerful

- Several outputs
- Professional quality
- ... and simple!

Markdown (II)

Markdown example

```
# I am a header
## I am a subheader

Regular, *italic* and **bold**

- List item 1
- List item 2

[I am a link](http://foo.com)

![I am a pic](markdown.png)

~~~C
printf("Hello, world");
~~~
```

I am a header

I am a subheader

Regular, *italic* and **bold**

- List item 1
- List item 2

I am a link

I am a pic

```C
printf("Hello, world");
```